MARINDUQUE STATE UNIVERSITY

COLLEGE OF INFORMATION AND COMPUTING SCIENCES

STUDENT RETENTION & ACADEMIC RISK EARLY WARNING SYSTEM (ISREMS)

Modules 1 to 3 Complete Database Schema & Data Dictionary (21 Tables)

PART 1: SYSTEM ARCHITECTURE & MODULE SCOPE OVERVIEW

This engineering specification details the complete relational database design for the first three (3) sub-modules of the MarSU Retention System (ISREMS): **(1) Student Information Management**, **(2) Academic Monitoring**, and **(3) Attendance and Engagement Monitoring**. While the centralized ERP core supplies reference tables, twenty-one (21) specialized normalized tables are required to store student demographic vulnerability risk variables, electronic class records, raw and transmuted assessment scores, roll-call sessions, compliance telemetry, and scholastic deficiencies.

PART 2: COMPLETE TABLE DIRECTORY MATRIX (21 TABLES)

#	Table Name	Module Scope	Entity Function & Storage Description
1	ret_student_demographics	Module 1: Profiles	Socio-economic, housing, commute risk, guardian reachability & 4Ps telemetry
2	ret_student_grades	Module 1: Profiles	Permanent semester academic transcripts, numerical grades, credited units, and flags
3	ret_student_enrollments	Module 1: Profiles	Matriculation header records per academic year and semester term
4	ret_student_study_load	Module 1: Profiles	Enrolled course subjects breakdown, units, and status per student per term
5	ret_subject_assignments	Module 2: Academic	Faculty instructor-to-subject course assignments, sections, and term offerings
6	ret_faculty_classes	Module 2: Academic	Electronic gradebook class records, cutoff policies, and class codes
7	ret_grade_components	Module 2: Academic	Class grading percentage weights (Quizzes, Performance Tasks, Periodic Exams)
8	ret_class_assessments	Module 2: Academic	Individual graded assessment activities, maximum points, and deadlines
9	ret_student_assessment_scores	Module 2: Academic	Raw student scores matrix per assessment item with feedback
10	ret_class_scores	Module 2: Academic	Transmuted periodic and final grades per student with risk status
11	ret_subject_performance_metrics	Module 2: Academic	Course-level passing rates, failure rates, bottleneck flags, and mean GPA
12	ret_academic_progress_reports	Module 2: Academic	Degree completion progress, earned units audit, and graduation milestones
13	ret_deficiencies	Module 2: Academic	Watchlist of failed (5.00), incomplete (INC), and dropped (DRP) courses
14	ret_deficiency_resolutions	Module 2: Academic	Removal examinations, compliance forms, and completion grade audit trail
15	ret_attendance_sessions	Module 3: Attendance	Class roll-call meeting headers with topics, meeting hours, and dates
16	ret_attendance_logs	Module 3: Attendance	Student daily presence status (Present, Absent, Late, Excused) and slips
17	ret_attendance_summaries	Module 3: Attendance	Cumulative term attendance rate and CHED 20% absence ceiling warning flags
18	ret_participation_tracking	Module 3: Attendance	Classroom recitation ratings, task submission rate, and quiz compliance
19	ret_student_engagement	Module 3: Attendance	Turnaround speed, LMS login telemetry, and overall engagement index
20	ret_behavior_records	Module 3: Attendance	Classroom conduct breaches, disciplinary cases, and intervention notes
21	ret_records	Core General Records	General administrative retention logs and operational records

PART 3: PHPMYADMIN COMPLETE DATA DICTIONARY SPECIFICATIONS

1 ret_student_demographics

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Module: Module 1: Student Information Management • Stores student socio-economic characteristics, housing, commute risk, and guardian contact profiles.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Demographic record ID	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Unique institutional student number	
living_arrangement	enum('With Family', 'Boarding House', 'Dormitory', 'Rented Apartment', 'Living Alone')		No	With Family			Current residential arrangement	
residence_type	enum('Permanent', 'Temporary / Relocated')		No	Permanent			Residence category during active school term	
address_barangay	varchar(100)		Yes	NULL			Barangay of residence	
address_municipality	varchar(100)		Yes	NULL			Municipality in Marinduque (e.g. Boac, Gasan)	
address_province	varchar(100)		No	Marinduque			Province of residence	
commute_distance_km	decimal(5,2)		No	0.00			Estimated travel distance to campus in kilometers	
primary_transport	enum('Walking', 'Tricycle', 'Jeepney', 'Motorcycle', 'Van / Bus', 'Private Vehicle')		No	Tricycle			Primary mode of transport to campus	
commute_risk_level	enum('Low (Local Resident)', 'Moderate (Relocated)', 'High (Distant Commute)')		No	Low (Local Resident)			Commuting vulnerability early warning level	
guardian_name	varchar(191)		Yes	NULL			Parent or legal guardian complete name	
guardian_relationship	varchar(50)		No	Parent			Relationship to student (Father, Mother, Relative)	
guardian_contact	varchar(50)		Yes	NULL			Emergency mobile contact number	
is_reachable	tinyint(1)		No	1			1 = Verified working line; 0 = Unreachable / No line	
family_monthly_income	enum('Below 10,000', '10,001 - 20,000', '20,001 - 40,000', 'Above 40,000')		No	10,001 - 20,000			Family income classification bracket	
is_4ps_beneficiary	tinyint(1)		No	0			1 = Pantawid Pamilyang Pilipino Program grantee	
is_working_student	tinyint(1)		No	0			1 = Working while studying (High attrition risk)	
is_solo_parent_household	tinyint(1)		No	0			1 = Single parent or self-supporting household	
has_home_internet	tinyint(1)		No	1			1 = Has broadband / fiber access; 0 = Mobile data only	
created_at	datetime		No				Record creation timestamp	
updated_at	datetime		Yes	NULL			Last profile update timestamp	
deleted_at	datetime		Yes	NULL			Soft deletion timestamp	

2 ret_student_grades

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 1: Student Information Management • Stores historical semester grades, transcript records, credited units, and prerequisite flags.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Grade record identifier	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
subject_id	int(11)		No			-> subjects.id ON DELETE CASCADE	Course curriculum subject ID	
subject_code	varchar(32)		No				Course subject code (e.g. CC111, IT211)	
subject_title	varchar(191)		No				Full official course descriptive title	
academic_year	varchar(32)		No	2025-2026			School academic year (e.g. 2025-2026)	
semester	varchar(32)		No	1st Semester			Semester term (1st Semester, 2nd Semester, Summer)	
year_level	tinyint(2)		No	1			Student curriculum year level when taken	
grade	varchar(10)		No				Official numerical grade (1.00 - 3.00, 5.00, INC, DRP)	
units	decimal(3,1)		No	3.0			Curricular units earned / credited	
remarks	enum('PASSED', 'FAILED', 'INCOMPLETE', 'DROPPED')		No	PASSED			Official academic grading remark	
curricular_flag	enum('Credited', 'Prerequisite Blocked', 'Removal Required', 'Retake Required')		No	Credited			Curricular trajectory warning flag	
instructor_id	int(11)		Yes	NULL		-> employees.id ON DELETE SET NULL	Faculty instructor who submitted final grade	
created_at	datetime		No				Grade submission timestamp	
updated_at	datetime		Yes	NULL			Grade revision timestamp	
deleted_at	datetime		Yes	NULL			Soft deletion timestamp	

3 ret_student_enrollments

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 1: Student Information Management • Tracks term-by-term matriculation registration headers, scholarships, and enrolled units.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Enrollment record ID	
enrollment_no	varchar(64)		No			UNIQUE KEY	Official registration transaction code (e.g. ENR-2026-101)	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
academic_year_id	int(11)		Yes	NULL		-> academic_years.id ON DELETE SET NULL	Academic year reference ID	
semester_id	int(11)		Yes	NULL		-> semesters.id ON DELETE SET NULL	Semester reference ID	
academic_year	varchar(32)		No	2026-2027			School academic year label	
semester	varchar(32)		No	1st Semester			Semester term label	
year_level	tinyint(2)		No	1			Year standing during registration	
section_id	int(11)		Yes	NULL		-> sections.id ON DELETE SET NULL	Assigned official class section	
program_id	int(11)		Yes	NULL		-> programs.id ON DELETE SET NULL	Degree program reference	
registration_status	enum('Officially Enrolled', 'Pending Assessment', 'Late Enrolled', 'Withdrawn', 'On Leave')		No	Officially Enrolled			Matriculation registration standing	
scholarship_status	varchar(100)		No	UniFAST Free Higher Education			Tuition subsidy / scholarship grant	
total_credit_units	decimal(4,1)		No	23.0			Sum of registered course units for the term	
enrolled_at	datetime		No				Official matriculation approval timestamp	
created_at	datetime		No				Record creation timestamp	
updated_at	datetime		Yes	NULL			Record update timestamp	

4 ret_student_study_load

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 1: Student Information Management • Stores line-item course subjects officially enrolled per student for the active term.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Study load item ID	
enrollment_id	int(11)		No			-> ret_student_enrollments.id ON DELETE CASCADE	Parent enrollment header reference	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
subject_id	int(11)		No			-> subjects.id ON DELETE CASCADE	Enrolled curriculum subject ID	
course_code	varchar(32)		No				Subject code (e.g. CC111, GE101)	
course_title	varchar(191)		No				Descriptive course title	
section_id	int(11)		Yes	NULL		-> sections.id ON DELETE SET NULL	Enrolled class section reference	
instructor_id	int(11)		Yes	NULL		-> employees.id ON DELETE SET NULL	Assigned faculty instructor	
lecture_hours	decimal(3,1)		No	3.0			Weekly lecture hours	
lab_hours	decimal(3,1)		No	0.0			Weekly laboratory hours	
units	decimal(3,1)		No	3.0			Credit units of the course	
status	enum('Official Enrolled', 'Added', 'Dropped', 'Withdrawn')		No	Official Enrolled			Study load enrollment status	
created_at	datetime		No				Study load encoding timestamp	
updated_at	datetime		Yes	NULL			Study load revision timestamp	

5 ret_subject_assignments

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Links employees (faculty members) to curriculum courses, sections, rooms, and terms.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Teaching assignment ID	
subject_id	int(11)		No			-> subjects.id ON DELETE CASCADE	Curriculum course subject reference	
employee_id	int(11)		No			-> employees.id ON DELETE CASCADE	Assigned faculty member reference	
section_id	int(11)		Yes	NULL		-> sections.id ON DELETE SET NULL	Assigned class section reference	
academic_year	varchar(32)		No	2026-2027			Active school year	
semester	varchar(32)		No	1st Semester			Active semester offering	
teaching_hours_per_week	decimal(4,1)		No	3.0			Total faculty contact hours per week	
room_id	int(11)		Yes	NULL		-> rooms.id ON DELETE SET NULL	Designated classroom / laboratory room	
schedule_day	varchar(50)		No	Monday / Wednesday			Assigned class meeting days	
schedule_time	varchar(50)		No	08:00 AM - 09:30 AM			Assigned class meeting schedule period	
status	enum('Active', 'Completed', 'Archived')		No	Active			Assignment offering lifecycle status	
created_at	datetime		No				Assignment entry timestamp	
updated_at	datetime		Yes	NULL			Assignment modification timestamp	
deleted_at	datetime		Yes	NULL			Soft deletion timestamp	

6 ret_faculty_classes

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Electronic gradebook class records, cutoff policies, and class codes created by faculty.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Faculty class record ID	
class_code	varchar(64)		No			UNIQUE KEY	System unique class record code (e.g. CLS-CC111-1001)	
faculty_id	int(11)		No			-> employees.id ON DELETE CASCADE	Instructor employee ID reference	
subject_id	int(11)		Yes	NULL		-> subjects.id ON DELETE SET NULL	Curriculum course ID reference	
course_code	varchar(32)		No				Subject code (e.g. CC111, ISP111)	
course_title	varchar(191)		No				Descriptive course title	
section	varchar(64)		No				Section name (e.g. BSIS 1A, BSTM 1st Year)	
school_year	varchar(32)		No	2026-2027			School academic year	
semester	varchar(32)		No	1st Semester			Semester term	
passing_cutoff	decimal(4,2)		No	3.00			Transmuted passing cutoff grade threshold	
grading_scheme	text		Yes	NULL			JSON configuration for grading components & percentage weights	
is_locked	tinyint(1)		No	0			1 = Class record finalized & submitted to Registrar; 0 = Open	
created_at	datetime		No				Class record creation timestamp	
updated_at	datetime		Yes	NULL			Class record modification timestamp	
deleted_at	datetime		Yes	NULL			Soft deletion timestamp	

7 ret_grade_components

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Configures grading criteria and percentage weights (Quizzes, Written Tasks, Midterm Exam).

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Grade component ID	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent faculty class record reference	
component_name	varchar(100)		No				Component label (e.g. Quizzes, Written Works, Exams)	
component_code	varchar(32)		No				Short identifier code (e.g. QZ, WW, PT, ME, FE)	
weight_percentage	decimal(5,2)		No	25.00			Weight percentage in final grade (e.g. 25.00%)	
order_index	tinyint(2)		No	1			Display sorting sequence order	
created_at	datetime		No				Creation timestamp	
updated_at	datetime		Yes	NULL			Update timestamp	

8 ret_class_assessments

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Stores individual assessment activities, maximum possible points, passing scores, and deadlines.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Assessment item ID	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent faculty class record reference	
component_id	int(11)		No			-> ret_grade_components.id ON DELETE CASCADE	Grading weight category reference	
title	varchar(191)		No				Assessment activity title (e.g. Quiz 1: Database Normalization)	
max_points	decimal(5,2)		No	50.00			Maximum total points attainable	
passing_score	decimal(5,2)		No	30.00			Base passing score threshold (60%)	
date_conducted	date		Yes	NULL			Date assessment was administered in class	
term_period	enum('Prelim', 'Midterm', 'Semi-Final', 'Final')		No	Midterm			Grading period period tier	
created_at	datetime		No				Activity creation timestamp	
updated_at	datetime		Yes	NULL			Activity update timestamp	

9 ret_student_assessment_scores

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Stores individual student raw scores, submission status, and faculty feedback per assessment.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Score entry ID	
assessment_id	int(11)		No			-> ret_class_assessments.id ON DELETE CASCADE	Assessment activity reference	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
raw_score	decimal(5,2)		No	0.00			Raw numeric points earned by student	
is_excused	tinyint(1)		No	0			1 = Officially excused by instructor; 0 = Normal	
is_late	tinyint(1)		No	0			1 = Submitted past deadline; 0 = On time	
feedback_notes	text		Yes	NULL			Faculty instructor diagnostic notes on student performance	
created_at	datetime		No				Score entry timestamp	
updated_at	datetime		Yes	NULL			Score revision timestamp	

10 ret_class_scores

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Stores student computed transmuted midterm and final semester grades with academic risk flags.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Class evaluation summary ID	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent class record reference	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Enrolled student university ID	
midterm_grade	decimal(4,2)		Yes	NULL			Computed transmuted midterm grade	
final_grade	decimal(4,2)		No	5.00			Computed transmuted final semester grade	
transmuted_rating	varchar(10)		No	5.00			University grade equivalent (1.00 - 3.00, 5.00, INC)	
status	enum('passed', 'failed', 'incomplete', 'dropped')		No	passed			Overall course passing status	
is_at_risk	tinyint(1)		No	0			1 = Student grade in danger zone (2.75 - 5.00)	
created_at	datetime		No				Record creation timestamp	
updated_at	datetime		Yes	NULL			Record update timestamp	

11 ret_subject_performance_metrics

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Aggregates course passing rates, failure rates, bottleneck flags, and student risk counts per term.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Analytics metric ID	
subject_id	int(11)		No			-> subjects.id ON DELETE CASCADE	Curriculum course subject reference	
academic_year	varchar(32)		No	2026-2027			Evaluation academic school year	
semester	varchar(32)		No	1st Semester			Evaluation semester term	
enrolled_count	int(11)		No	0			Total student cohort enrolled across all sections	
passed_count	int(11)		No	0			Total students who passed the course	
failed_count	int(11)		No	0			Total students who failed (5.00)	
dropped_count	int(11)		No	0			Total students who officially dropped (DRP)	
passing_rate	decimal(5,2)		No	0.00			Cohort passing percentage rate	
failure_rate	decimal(5,2)		No	0.00			Cohort failure percentage rate	
mean_gpa	decimal(4,2)		No	2.00			Average cohort grade point average	
is_bottleneck	tinyint(1)		No	0			1 = Curricular bottleneck subject (Passing rate < 85%)	
computed_at	datetime		No				Telemetry calculation timestamp	

12 ret_academic_progress_reports

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Tracks student cumulative degree progress, earned units against curriculum, and graduation targets.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Progress report ID	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
program_id	int(11)		No			-> programs.id ON DELETE CASCADE	Degree program reference	
curriculum_year	varchar(32)		No	2026-2027			Applicable CHED prospectus curriculum year	
total_required_units	decimal(5,1)		No	144.0			Total required degree units for graduation	
earned_units	decimal(5,1)		No	0.0			Total credited units successfully completed	
remaining_units	decimal(5,1)		No	144.0			Remaining uncompleted curricular units	
percent_progress	decimal(5,2)		No	0.00			Curricular completion percentage rate	
cumulative_gwa	decimal(4,2)		No	2.00			Overall general weighted average	
milestone_status	enum('On-Track', 'Delayed', 'Critical / MRR Risk', 'Graduating Candidate')		No	On-Track			Retention milestone progress status	
target_graduation_year	int(4)		No	2030			Projected academic graduation year	
evaluation_date	date		No				Official evaluation audit date	
created_at	datetime		No				Report generation timestamp	
updated_at	datetime		Yes	NULL			Report update timestamp	

13 ret_deficiencies

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Tracks failed grades (5.00), incomplete marks (INC), and dropped courses with regulatory deadlines.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Deficiency record identifier	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Deficient student identity number	
subject_id	int(11)		No			-> subjects.id ON DELETE CASCADE	Deficient subject course identifier	
employee_id	int(11)		Yes	NULL		-> employees.id ON DELETE SET NULL	Instructor responsible for completion exam	
deficiency_type	enum('Failed (5.00)', 'Incomplete (INC)', 'Dropped (DRP)')		No				Scholastic deficiency category	
grade_value	varchar(10)		No	INC			Recorded deficient grade mark	
academic_year	varchar(32)		No	2026-2027			Academic school year when incurred	
semester	varchar(32)		No	1st Semester			Academic semester when incurred	
deadline_date	date		Yes	NULL			Statutory university deadline for compliance (1 year)	
status	enum('Pending', 'In Progress', 'Resolved', 'Lapsed')		No	Pending			Deficiency lifecycle resolution status	
created_at	datetime		No				Deficiency identification timestamp	
updated_at	datetime		Yes	NULL			Resolution update timestamp	

14 ret_deficiency_resolutions

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 2: Academic Monitoring • Audits deficiency removal exams, fulfillment requirements, and official completion grade slips.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Resolution record identifier	
deficiency_id	int(11)		No			-> ret_deficiencies.id ON DELETE CASCADE	Parent deficiency reference ID	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
removal_exam_date	date		Yes	NULL			Date completion exam was taken	
requirements_fulfilled	text		Yes	NULL			Summary of academic requirements submitted	
completion_grade	decimal(4,2)		Yes	NULL			Final transmuted completion grade (1.00 - 3.00, or 5.00)	
faculty_signature_date	date		Yes	NULL			Date signed by course instructor	
registrar_official_id	int(11)		Yes	NULL		-> users.id ON DELETE SET NULL	University Registrar validator user ID	
official_receipt_no	varchar(64)		Yes	NULL			Cashier official receipt number for completion exam fee	
created_at	datetime		No				Resolution timestamp	

15 ret_attendance_sessions

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Stores class roll-call meeting headers, lecture topics, meeting hours, and proctors.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Attendance session identifier	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent faculty class record ID	
subject_id	int(11)		Yes	NULL		-> subjects.id ON DELETE SET NULL	Curriculum course ID reference	
session_date	date		No				Date of class meeting session	
session_type	enum('Lecture', 'Laboratory', 'Fieldwork', 'Online Virtual')		No	Lecture			Instructional meeting mode	
start_time	time		Yes	NULL			Class session start time	
end_time	time		Yes	NULL			Class session end time	
topic_covered	varchar(255)		Yes	NULL			Syllabus topic discussed during class	
recorded_by	int(11)		Yes	NULL		-> users.id ON DELETE SET NULL	User account who logged the roll call	
created_at	datetime		No				Session logging timestamp	

16 ret_attendance_logs

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Stores individual student daily attendance marks (Present, Absent, Late, Excused) and excuse slips.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Student attendance log identifier	
session_id	int(11)		No			-> ret_attendance_sessions.id ON DELETE CASCADE	Parent session meeting ID	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
status	enum('Present', 'Absent', 'Excused', 'Late')		No	Present			Daily presence status	
time_in	time		Yes	NULL			Timestamp of student arrival in classroom	
excuse_slip_path	varchar(255)		Yes	NULL			File path to approved clinic / guidance excuse slip	
remarks	varchar(255)		Yes	NULL			Reason or notes for absence / tardiness	
created_at	datetime		No				Log entry timestamp	

17 ret_attendance_summaries

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Stores aggregated term attendance rates and CHED 20% absence threshold breach warnings.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Summary telemetry ID	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent class record reference	
academic_year	varchar(32)		No	2026-2027			School academic year	
semester	varchar(32)		No	1st Semester			Semester term	
total_sessions	int(11)		No	30			Total class sessions held to date	
present_count	int(11)		No	0			Total sessions student was present	
absent_count	int(11)		No	0			Total unexcused absences incurred	
excused_count	int(11)		No	0			Total excused medical/official absences	
late_count	int(11)		No	0			Total tardiness incidents	
attendance_rate	decimal(5,2)		No	100.00			Cumulative presence percentage rate	
consecutive_absences	int(11)		No	0			Consecutive unexcused absences (Risk flag trigger at 3)	
threshold_breached	tinyint(1)		No	0			1 = Breached CHED 20% absence limit (Automatic DRP/Failure risk)	
last_calculated_at	datetime		No				Telemetry calculation timestamp	

18 ret_participation_tracking

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Tracks student classroom task compliance, recitation ratings, and missed assignment metrics.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Participation record identifier	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
class_id	int(11)		No			-> ret_faculty_classes.id ON DELETE CASCADE	Parent class record reference	
quizzes_done	int(11)		No	0			Total class quizzes completed	
quizzes_total	int(11)		No	8			Total class quizzes scheduled	
activities_done	int(11)		No	0			Total lab & practical activities completed	
activities_total	int(11)		No	12			Total lab & practical activities scheduled	
missed_tasks	int(11)		No	0			Total unsubmitted / delinquent tasks	
compliance_rate	decimal(5,2)		No	100.00			Task submission compliance percentage	
recitation_rating	enum('Exemplary', 'Satisfactory', 'Needs Improvement', 'Passive')		No	Satisfactory			Classroom recitation / engagement qualitative mark	
teacher_comments	text		Yes	NULL			Instructor qualitative remarks on student engagement	
updated_at	datetime		No				Participation update timestamp	

19 ret_student_engagement

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Tracks learning turnaround speed, LMS access telemetry, and overall student engagement index.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Engagement record identifier	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student university ID number	
academic_year	varchar(32)		No	2026-2027			School academic year	
semester	varchar(32)		No	1st Semester			Semester term	
turnaround_speed	enum('Early Turn-in', 'On-Time', 'Late (-1 Day)', 'Late (-2 Days)', 'Chronic Delinquent')		No	On-Time			Punctuality index in submitting coursework	
engagement_score	decimal(5,2)		No	85.00			Composite retention engagement index score (0 - 100)	
lms_logins_per_week	int(11)		No	5			Average university portal logins per week	
resource_downloads	int(11)		No	12			Total syllabus modules and lecture slides accessed	
engagement_tier	enum('High Engagement', 'Moderate Compliance', 'Disengaged / Passive', 'Severe Academic Disconnection')		No	Moderate Compliance			Qualitative behavioral engagement tier	
last_activity_at	datetime		Yes	NULL			Timestamp of last verified academic interaction	
created_at	datetime		No				Creation timestamp	
updated_at	datetime		Yes	NULL			Telemetry update timestamp	

20 ret_behavior_records

Creation: Oct 16, 2026 at 09:00 AM

Module: Module 3: Attendance and Engagement Monitoring • Records student disciplinary incidents, classroom conduct breaches, and behavioral risk flags.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: Incident log identifier	
student_id	varchar(64)		No			-> students.student_number ON DELETE CASCADE	Student involved in conduct incident	
incident_type	varchar(191)		No				Category of incident (e.g. Tardiness, Classroom Conduct, Dishonesty)	
description	text		Yes	NULL			Factual narrative of conduct breach	
severity	enum('Minor', 'Moderate', 'Severe')		No	Minor			Disciplinary violation severity level	
reported_by	int(11)		Yes	NULL		-> users.id ON DELETE SET NULL	User account who reported incident	
incident_date	date		No				Date incident occurred	
action_taken	text		Yes	NULL			Corrective intervention or counseling reprimand applied	
status	enum('Reported', 'Under Investigation', 'Resolved', 'Dismissed')		No	Reported			Disciplinary case lifecycle status	
created_at	datetime		No				Incident logging timestamp	
updated_at	datetime		Yes	NULL			Incident update timestamp	

21 ret_records

Creation: Oct 16, 2026 at 09:00 AM

Module: Core Retention General Records Table • General administrative retention logs and operational records.

Column	Type	Attributes	Null	Default	Extra	Links to	Comments	MIME
id	int(11)		No		auto_increment		Primary Key: General record identifier	
title	varchar(191)		No				General log entry title	
description	text		Yes	NULL			Detailed narrative description	
category	varchar(100)		No	General			Administrative retention category	
status	enum('active', 'pending', 'resolved', 'archived')		No	active			General lifecycle status	
created_by	int(11)		Yes	NULL		-> users.id ON DELETE SET NULL	Account who filed the record	
created_at	datetime		No				Entry filing timestamp	
updated_at	datetime		Yes	NULL			Entry modification timestamp	
deleted_at	datetime		Yes	NULL			Soft deletion timestamp	

PART 4: RELATIONAL INTEGRITY & FOREIGN KEY ARCHITECTURE

All 21 ISREMS retention tables strictly interconnect with baseline core tables using non-destructive Foreign Key constraints. Foreign keys referencing `students.student_number` enforce `ON DELETE CASCADE` to preserve referential consistency, while links to `employees.id` and `users.id` utilize `ON DELETE SET NULL` to preserve academic audit histories.